Mohamed Elhadad

.NET Full Stack Developer

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PROFESSIONAL SUMMARY

Dedicated .NET Full Stack Developer and Computer Science student with a strong foundation in C#, ASP.NET Core, and Microsoft SQL Server. Experienced in implementing Clean Architecture, multi-tier systems, relational database design, and real-time communication with SignalR. Committed to software quality with over 500 automated tests written across projects, with expertise in handling concurrency, race conditions, and secure RESTful APIs.

TECHNICAL SKILLS

Backend: C#, .NET 8 / 9 / 10, ASP.NET Core MVC, Web API, Minimal APIs, Entity Framework Core, LINQ, SignalR

Database & Persistence: Microsoft SQL Server, PostgreSQL, SQLite, Relational Modeling, Migrations, Indexing, sp_getapplock Locking

Frontend & Web: HTML5, CSS3, JavaScript (ES6+), Bootstrap 5, Leaflet.js Maps, PWA, Responsive UI

Architecture & Patterns: Clean Architecture, N-Tier, Repository & Unit of Work, Dependency Injection, RESTful APIs, OOP, Design Patterns

Testing & Quality: xUnit, Moq, FluentAssertions, Concurrency Stress Tests, Integration Testing (500+ Automated Tests)

DevOps & Tools: Git, GitHub, GitHub Actions (CI/CD), Docker, Docker Compose, Swagger / OpenAPI 3.0, Postman

FEATURED ENGINEERING PROJECTS

CourtBook (PlaySpot Web Portal) — Sports Facility Booking Platform

.NET 10 · Clean Architecture

C# 13, ASP.NET Core, EF Core, SQL Server, SignalR, Paymob API, Docker, xUnit (362 tests)

- Architected enterprise court booking system following strict Clean Architecture (Domain, Application, Infrastructure, REST API, Razor Pages Web).
- Eliminated double-booking race conditions during high-volume slot bookings using SQL Server `sp_getapplock` distributed locking.
- Authored **362 automated unit, integration, and concurrency stress tests** validating availability, promo codes, and pricing rules.
- Integrated real-time SignalR WebSocket hubs for instant court availability and Paymob payment API (cards & mobile wallets).

MediCare — Clinic Management & Appointment System

.NET 8 · DEPI Graduation Project

C#, ASP.NET Core MVC, EF Core, SQL Server, SignalR, xUnit (157 tests), Identity

- Developed multi-role clinic management solution (Administrator, Doctor, Patient) following N-Tier architecture.
- Guaranteed appointment booking integrity and concurrency protection via SQL Server **filtered unique indexes**.
- Built comprehensive clinical records, electronic prescription workflows, and automated schedule conflict management.
- Maintained **157 automated tests** with Moq and FluentAssertions verifying business logic and authorization policies.

UniNest — Full-Stack Student Housing Platform

.NET 9 · Web API & PWA

C#, Web API, Clean Architecture, EF Core, SQL Server / SQLite, Leaflet.js, PWA

- Engineered student housing discovery REST API with JWT Bearer authentication, role-based access, and 120 req/min rate limiting.
- Implemented provider-agnostic persistence supporting both Microsoft SQL Server and portable SQLite fallback.
- Built interactive PWA frontend featuring Leaflet.js university proximity maps, live search, student reviews, and comparison tools.

EDUCATION & TRAINING

Bachelor of Science in Computer Science — Egyptian E-Learning University (EELU) - Fayoum — Expected 2026

.NET Full-Stack Web Development Track — Digital Egypt Pioneers Initiative (DEPI) 2026

CERTIFICATIONS & MILESTONES

Azure Fundamentals (AZ-900) — Microsoft August 2026

CCNA: Introduction to Networks — Cisco Networking Academy · EELU May 2026

Generative AI Certificate — CIB Summer Internship Program July 2026

International Certification for Artificial Intelligence License — AMG Academy 2025